going out of sample. Consistent with the spurious high R^2s, Welch and Goyal (2008) find that the historical average of excess stock returns forecasts better than almost all predictive variables. Use smart econometric techniques that combine a lot of information, but be careful about data mining and take into account the possibility of shifts in regime.[29]

2. Use economic models.

   Notice that the best predictors in Table 8.7 were valuation ratios. Prediction of equity risk premiums is the same as prediction of economic value. If you can impose economic structure, do it. Campbell and Thompson (2008), among others, find that imposing economic intuition and constraints from economic models helps.[30]

3. Be humble.

   If you're trying to time the market, then have humility. Predicting returns is hard. Since it is difficult to statistically detect predictability, it will also be easy to delude yourself into thinking you are the greatest manager in the world because of a lucky streak (this is *self-attribution bias*), and this overconfidence will really hurt when the luck runs out. You will also need the right governance structure to withstand painful periods that may extend for years (see chapter 15). Asset owners are warned that there are very few managers who have skill, especially among those who think they have skill (see also chapters 17 and 18 on hedge funds and private equity, respectively).

Since the predictability in data is weak, investors are well served by taking the i.i.d. environment as a base case. This means dynamic asset allocation with rebalancing to constant weights (or exposures) along the lines of chapter 4. If there is some mean reversion of returns in the data, as suggested by the low returns that follow high prices, and vice versa, then rebalancing back to constant weights will be advantageous. Recall that rebalancing is a counter-cyclical strategy; it buys when equities have fallen in price, so the equity position is increased when expected returns are high and sells when equities have risen in price, so the equity position is reduced when expected returns are low. You can do better than this if you can predict the future, but you probably can't. At least rebalancing will get you going in the right direction—the truly optimal (ideal) strategy will only be *more* aggressive than rebalancing. If you can't do constant rebalancing, then you are highly unlikely to be able to undertake the optimal investment strategy when the true data-generating process of the equity risky premium exhibits predictability.

[29] See Timmermann (2006) for a summary on optimal forecast combination.

[30] Ang and Piazzesi (2003) impose no-arbitrage constraints and find these improve forecasts of interest rates. Campbell and Thompson (2008) impose that forecasts of equity risk premiums are positive by truncating at zero. A more general approach is Pettenuzzo, Timmermann, and Valkanov (2013), who impose bounds on Sharpe ratios.